

shaded circles

R_1 at T_2 R_3 at T_3

$$N(C_3 C_4) = 3 \times 3!$$

$$N(C_2 C_4) = 3 \times 3!$$

$$N(C_2 C_3) = 3 \times 3!$$

$$N(C_1 C_2 C_3) = 2 \times 2!$$

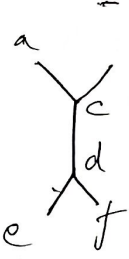
$$A. = 25.$$

$$1 + 7n + 16n^2 + 13n^3 + 3n^4$$

UNIT V - GRAPHS

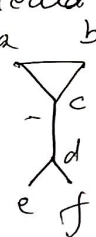
13/12/17
Wednesday

trees - no cycles
- connected



41

29



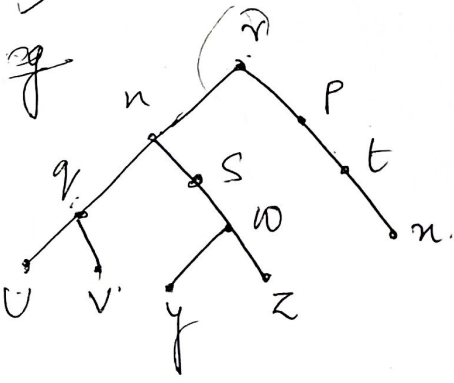
42

X



43

forest (2 trees)



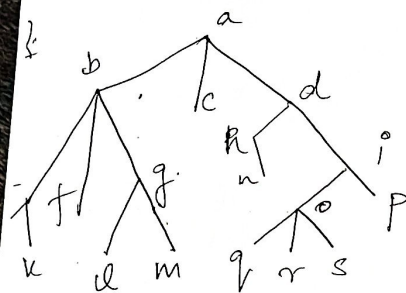
a graph G is called as a rooted tree if there is a unique vertex r called as root in G with $\text{in degree}(r) = 0$ and $\forall$ other vertices v , $\text{in degree}(v) = 1$

$\text{out degree} = 0 \Rightarrow \text{leaf nodes}$

level (s) = path from r (root) to s = 2.

ancestors of w, y, z = s, u, r.

descendants s, u, r = w, y, z



(i) which vertex is root a

(ii) which vertices are internal = a, b, c, d, h, i, o.
⇒ non leaf nodes

(iii) leaf = j, k, l, m, c, n, q, r, s, p

(iv) children (i) = o, p.

(v) parent (u) = d.

(vi) sibling (o) = p.

(vii) ancestors (u)

= g, b, a

(viii) descendants (b) = e, j, k, f, g, l, m

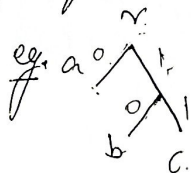
(ix) level (f) = 2.

(x) at level 4 = q, r, s.

⇒ ^{LDP} inorder, ^{DLP} preorder, ^{LDP} postorder

Optimal prefix code

prefix code - set P of binary sequence s is called as a prefix code if no sequence in P is a prefix of any other sequence in P.



a: 0
b: 10
c: 11

P = {0, 10, 11}

(child = 0

child = 1

0 is not prefix of 10 or 11

a: 01 e: 0 n: 1101 r: 10 t: 1

ata : 01101 or 01101
 $\begin{array}{c} \downarrow \downarrow \downarrow \\ e \quad n \end{array}$ $\begin{array}{c} \downarrow \downarrow \downarrow \downarrow \\ e \quad t \quad r \quad e \end{array}$

aka
 ef

∴ not prefix code. (ata can be encoded as en or etrt)

Q. a: 111 e: 0 n: 1100 r: 1101 t: 10

ata : 11110111 ✓
 $\begin{array}{c} \downarrow \downarrow \downarrow \downarrow \\ a \quad t \quad r \quad a \end{array}$

A = 01

B = 011

∴ no other combination is possible. c = 11

Huffman's tree

Step 1: Assign unique weights one each to set S of n isolated vertices. while $|S| > 1$ then perform the following:-

- ① find 2 trees T and T' in S with smallest root weights w and w' respectively.
- ② Create a new complete binary tree T* with root weight as $w^* = w + w'$ and having T and T' as its left & right subtrees respectively.
- ③ Place T* in S and delete T and T'

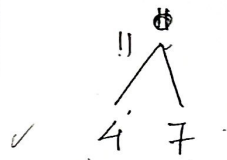
3 marks

Q. Construct an optimal prefix code for the symbols, a, o, q, v, y, z that occur with the frequencies 20, 28, 4, 17, 12, 7

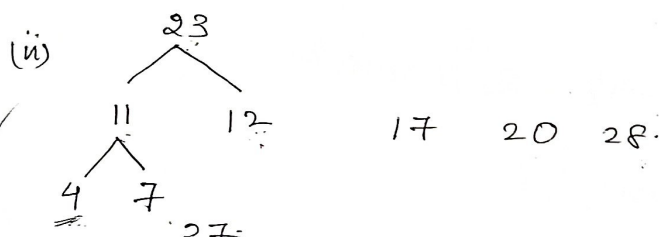
⇒ if frequency is not given, then a string will be given

4. (i) array with in ascending order.

② 4 7 12 17 20 28.

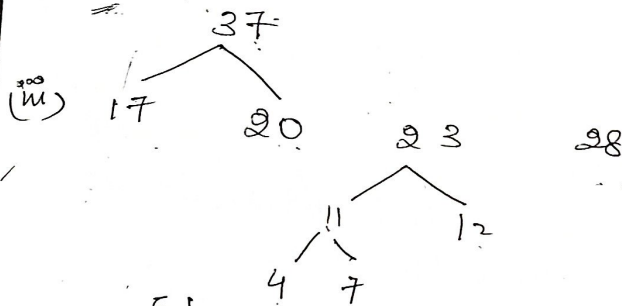


10 marks

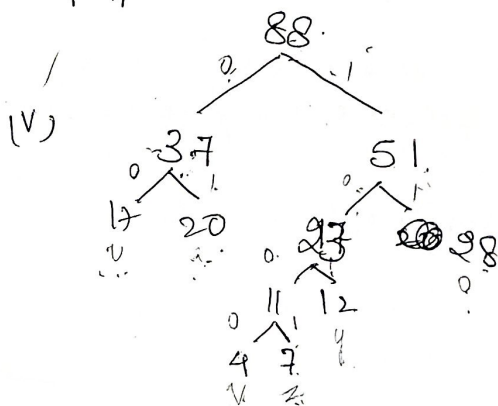
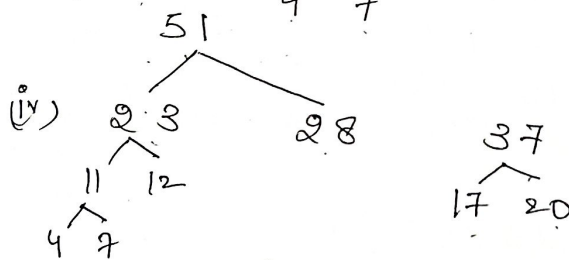


Engineering

E-2



1 2 2 4 5



a: 01
 u: 00
 o: 11
 y: 101
 z: 1001
 q: 1000

Construct an array with

(ii)

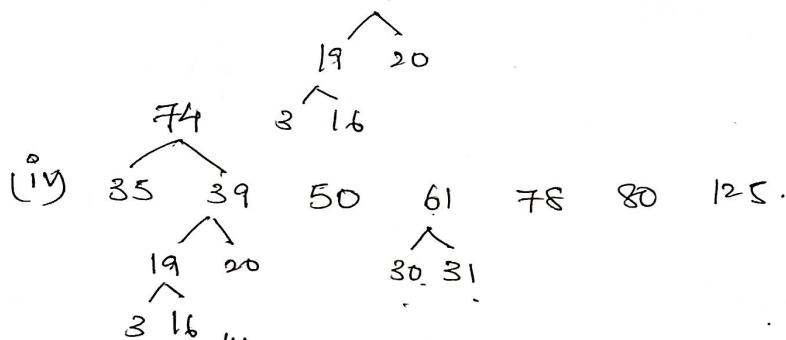
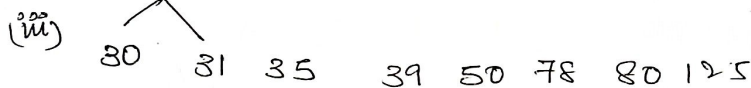
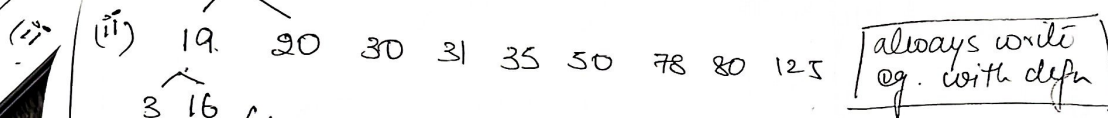
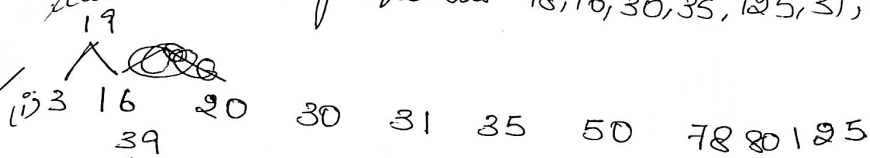
(ii)

(i)

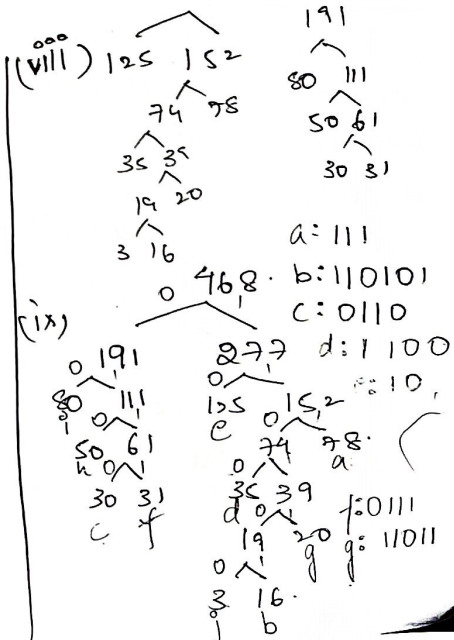
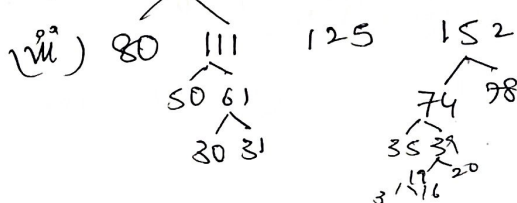
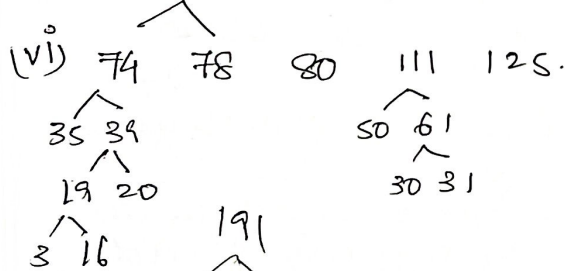
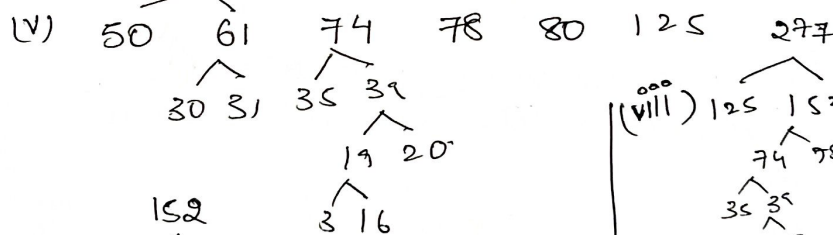
(i)

(v)

Construct an op. prefix code for symbols a to j that occur with the frequencies 78, 16, 30, 35, 125, 31, 20, 50, 80, 3



$k: 010$
 $i: 00$
 $j: 110100$



Q Find the value of n, y & z :

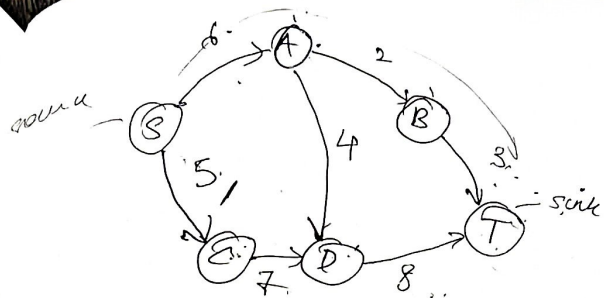
$a: 00$
 $b: 01$
 $c: 101$
 $d: n10$ A. (i) if $n=0$, $\begin{matrix} 010 \\ b \end{matrix} \times$
 $e: yz$ if $n=1$, $110 \checkmark$

(ii) $yz=00$, $\begin{matrix} 001 \\ a \end{matrix} \times$
 $yz=01$, $\begin{matrix} 011 \\ b \end{matrix} \times$
 $yz=10$, $\begin{matrix} 101 \\ c \end{matrix} \times$
 $yz=11$, $111 \checkmark$

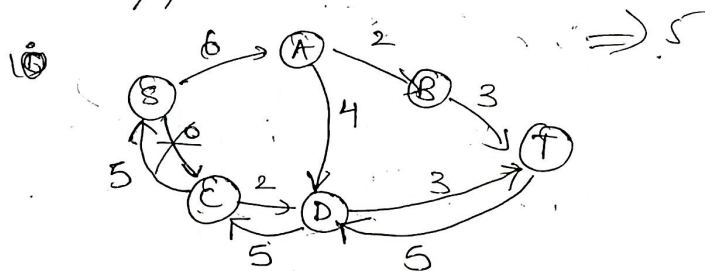
$\therefore n=y=z=1$

Ford Fulkerson

- ① If $\exists$ a path from source to sink, go to step ② otherwise the optimal solution is determined by the backward arcs.
- ② Calculate the residual in flow network,
 - (i) find the smallest capacity of the arcs $\Rightarrow$ denoted by c
 - (ii) subtract c from the capacity of each arc on the path
 - (iii) add a backward edge for each arc on the path then add c to the capacity of each reverse arc & go back to ①

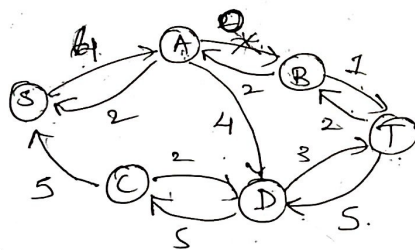


(i) choosing path $S \rightarrow C \rightarrow D \rightarrow T$

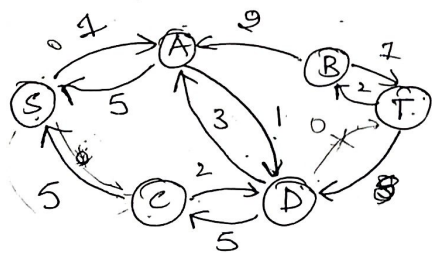


cycle $5+2$

(ii) choosing $S \rightarrow A \rightarrow B \rightarrow T$



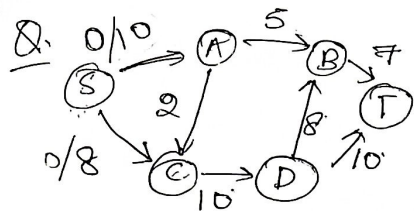
(iii) $S \rightarrow A \rightarrow D \rightarrow T$



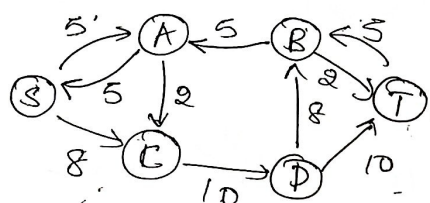
now no more paths possible.

$$\text{maxi num flow} = \underline{5} + \underline{2} + \underline{3} = \underline{10}$$

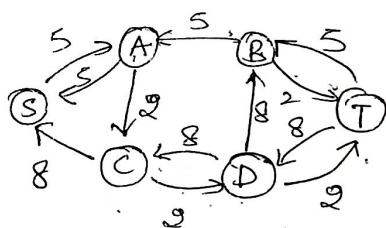
5 min capacity in all paths chosen.



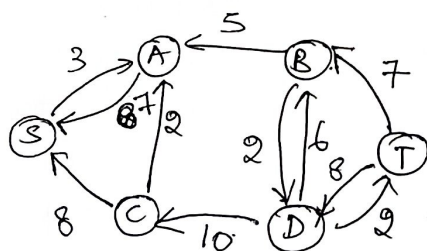
i) $S \rightarrow A \rightarrow B \rightarrow T$ 5



ii) $S \rightarrow C \rightarrow D \rightarrow T$ 8

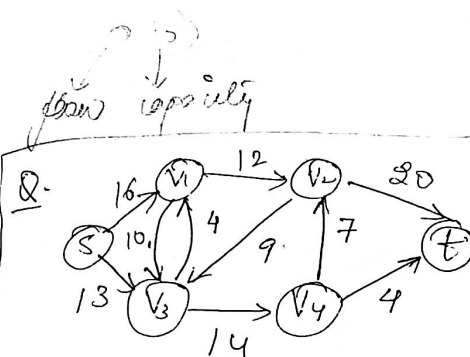


iii) $S \rightarrow A \rightarrow B \rightarrow T$ 2

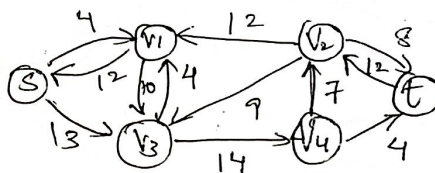


no more paths.

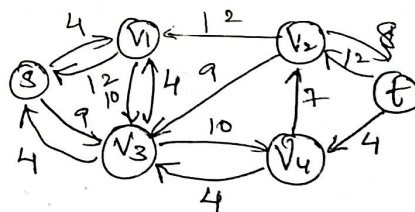
$$\text{max. flow} = 5 + 8 + 2 = 15$$



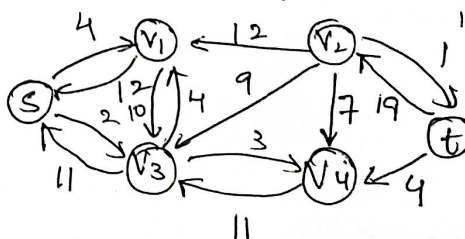
i) $S \rightarrow V_1 \rightarrow V_2 \rightarrow T$ 12



ii) $S \rightarrow V_3 \rightarrow V_4 \rightarrow T$ 4



iii) $S \rightarrow V_3 \rightarrow V_4 \rightarrow V_2 \rightarrow T$ 7



$$\begin{aligned} \text{max flow} &= 12 + 4 + 7 \\ &= 23 \end{aligned}$$